

Group Project: Mobile Application Development with Cloud Integration

Course: CSP650 (Example) - Mobile Application Development

Project Type: Group Project (4-5 members)

Submission: Final Report (PDF) & Video Presentation

1. Introduction & Objectives

This group project is the main practical component of this course. The primary goal is to design, develop, and deploy a fully functional mobile application that solves a real-world problem. The application must leverage modern cloud computing services for its backend infrastructure and business logic.

Learning Objectives:

- To gain practical experience in the complete mobile application development lifecycle.
- To design and implement a scalable, cloud-native application architecture.
- To integrate and manage multiple third-party cloud computing services (PaaS/SaaS/IaaS).
- To design, build, and deploy a custom "in-house" web service (API) to support the mobile application.
- To practice project management, teamwork, and technical documentation skills.

2. Project Task

Your group is tasked to propose, design, develop, and document a **mobile application** for either Android or iOS (or cross-platform using frameworks like Flutter or React Native).

The core of this project lies in the architecture. The application **must not** be a simple standalone app. It must be a cloud-integrated solution that meets the specific technical requirements below.

3. Technical Requirements

Your project submission will be evaluated on the successful implementation of the following three requirements:

a. Mobile Application (Android/iOS)

You must develop a native or cross-platform mobile application. Your group will define the problem statement, target audience, and key features of the app.

b. Integration with Third-Party Cloud Services (Minimum 3)

Your application must be integrated with **at least THREE (3)** different third-party cloud computing services. These services should provide the foundational backend for your app.

Examples of services you can use:

- **Authentication:**
 - Firebase Authentication (Google)
 - AWS Cognito (Amazon)
 - Auth0
 - Google Sign-In / Apple Sign-In
- **Database (DBaaS):**
 - Firebase Firestore / Realtime Database (Google)
 - AWS DynamoDB (Amazon)
 - MongoDB Atlas
- **Storage (Object Storage):**
 - Firebase Cloud Storage (Google)
 - AWS S3 (Amazon)
- **Location & Maps:**
 - Google Maps Platform (e.g., Maps SDK, Geocoding API)
 - Mapbox
- **Serverless Functions (FaaS):**
 - Cloud Functions for Firebase (Google)
 - AWS Lambda (Amazon)
- **Analytics:**
 - Google Analytics for Firebase
- **Machine Learning:**
 - Google ML Kit (on-device or cloud)
 - AWS Rekognition (for image analysis)
- **Notifications:**
 - Firebase Cloud Messaging (FCM)
 - Amazon Simple Notification Service (SNS)

c. Integration with In-House Web Service (Minimum 1)

In addition to the third-party services, your application must communicate with **at least ONE (1)** custom "in-house" web service.

- **What this means:** You must design, develop, deploy, and host your own backend RESTful API.
- **Purpose:** This API should provide custom business logic that is specific to your application and not available from a third-party service. (e.g., a custom recommendation engine, complex data processing, managing specific user logic).
- **Technology:** You can use any backend technology (e.g., Node.js/Express, Python/Flask/Django, Java/Spring Boot).
- **Hosting:** This web service must be deployed to a cloud platform (e.g., Heroku, Vercel, Google App Engine, AWS Elastic Beanstalk).

4. Recommended Methodology

While you are free to choose your project management methodology, we strongly recommend an **Agile approach (like Scrum)**. This involves working in "sprints" (e.g., 1-2 week cycles) to complete specific features.

Suggested Project Phases:

1. Phase 1: Planning & Design (Weeks 1-3)

- **Ideation:** Define the app's problem statement, target users, and core features.
- **Tech Stack:** Decide on your mobile framework (Native, Flutter, React Native), your 3+ third-party services, and your in-house API tech.
- **Architecture:** Create a detailed system architecture diagram showing how the mobile app, third-party services, and your in-house API connect.
- **Design:**
 - Create UI/UX wireframes and mockups (use tools like Figma or Balsamiq).
 - Design your database schema (e.g., your Firestore collections or SQL tables).
 - Define the API contract (endpoints, request/response formats) for your in-house web service.

2. Phase 2: Development & Integration (Weeks 4-10)

- Set up your cloud projects (Firebase, AWS, etc.) and development environments.
- **Backend First:** Develop and deploy the first version of your in-house web service.
- **Mobile App:** Begin building the mobile app UI.
- **Integrate:**
 - Connect the app to cloud authentication.
 - Connect the app to the cloud database and storage.
 - Integrate your in-house web service.
 - Integrate all other cloud services.
- **Iterate:** Work in sprints, testing features as you build them.

3. Phase 3: Testing & Deployment (Weeks 11-12)

- Perform thorough testing: unit testing, integration testing, and user acceptance testing (UAT).
- Fix bugs and refine the UI/UX.
- Ensure your in-house web service is stable and deployed.
- Prepare the final mobile application build (e.g., APK file).

4. Phase 4: Documentation & Presentation (Weeks 13-14)

- Write the final project report (see structure below).
- Record and edit your video presentation.

5. Deliverables

A. Final Report (PDF)

You must submit a formal project report. Your report **must** follow this exact structure:

1. **Cover page** (Course, Project Title, Group Members, Student IDs)
2. **Table of contents**
3. **Introduction** (Project background, problem statement, objectives, project scope)
4. **Cloud services used**
 - Detailed explanation of all 3+ third-party services used.
 - Detailed explanation of your 1+ in-house web service.
 - **Justification:** For *each* service, explain *what* it does and *why* you chose it over alternatives.
5. **Design**
 - **System Architecture:** A clear, detailed diagram.
 - **Database Design:** Schema diagrams (for Firestore, NoSQL, or SQL).
 - **Service Design:** API endpoint documentation for your in-house service.
 - **UI/UX Design:** Include key wireframes and mockups.
6. **Methodology**
 - Explain the development methodology you used (e.g., Agile, Waterfall) and how your team applied it (e.g., sprint planning, tools used like Trello/Jira). This is a critical section.
7. **Screenshots of your application**
 - High-quality screenshots that demonstrate all the key features and workflows of your application.
8. **Conclusion, limitation, and recommendation**
 - **Conclusion:** Summarize the project and what you achieved.
 - **Limitations:** Discuss any technical challenges, unmet goals, or limitations of your app.
 - **Recommendation:** Propose future work and improvements.
9. **References** (Use APA or IEEE format)
10. **Link to Video presentation** (Must be a shareable, working link, e.g., YouTube or Google Drive)

B. Video Presentation

You must create a video presentation of your project.

- **Length:** Maximum 15 minutes.
- **Content:**
 - Brief introduction of the team and the project.
 - Live demonstration of the final, working mobile application. This is the most important part.
 - Brief overview of your cloud architecture.
 - Discussion of challenges and lessons learned.
- **Link:** The link must be included in the final report.

C. Source Code

Submit a link to a private LINK (will be provided) containing the complete source code for:

1. The mobile application.
2. The in-house web service.